Fed-UGI: Federated Undersampling Learning Framework With Gini Impurity for Imbalanced Network Intrusion Detection

Ming Zheng, *Member, IEEE*, Xiaowen Hu, Ying Hu, Xiaoyao Zheng, *Senior Member, IEEE*, and Yonglong Luo

***Abstract*—In the modern interconnected world, the popularization of networks and the rapid development of information technology led to the increasing security risks and threats in network systems. The existing intrusion detection system is constantly challenged by various malicious intrusion attacks. Machine learning algorithms have been widely used in intrusion detection. However, the model training requires the support of a sufficient high-quality samples, especially attack traffic data. Network intrusion detection datasets may not be shared between organizations due to data security and some privacy policy concerns. The federated learning framework is an optimal approach to address this issue, in which organizations collaborate to train a global model shared by multiple parties while keeping the data local to the client, guaranteeing the data privacy and security of all parties. However, there is a problem of class imbalance in the network traffic data owned by the organizations, which seriously affects the detection performance of the model and leads to a high consumption of model training time. Therefore, this study proposed a novel federated undersampling learning framework with Gini impurity, namely Fed-UGI. The framework is based on the hash-based block undersampling method to rebalance the client, which can solve the influence of imbalanced training data on the model detection performance and improve the model training efficiency. Moreover, the client weighted aggregation strategy based on Local Gini impurity can further optimize the effect of global model aggregation and reduce the impact of the dispersion degree and information difference in client data on model aggregation. In addition, extensive experiments on intrusion detection datasets show that compared to SOTA methods, the proposed Fed-UGI method has a good detection effect on the three metrics of F1-score, G-mean and AUC, the training time of the model is reduced by 51.76%-92.58%, especially in highly class imbalance situation.**

***Index Terms*—Federated learning, imbalanced data, network intrusion detection, undersampling.**

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I. Introduction

With the further internationalization, openness and personalization of the Internet, the network makes our lives more convenient, but it also makes the threat of network security quite common in our daily lives. Criminals can take advantage of security loopholes in the network to carry out malicious activities such as illegal invasion, destruction, theft and tampering of target systems [1]. Illegal access to personal information, online fraud, network intrusion, cyber theft and other activities may lead to personal financial losses, privacy leakage, system paralysis, seriously affecting social production and life, and harming national interests. Network security is related to personal security, enterprise security, but also related to national security. In this case, further research is needed to establish a Network Intrusion Detection System (NIDS) that can effectively detect attacks to ensure a secure network environment [2].

In the past decade, various network intrusion detection methods based on machine learning have been proposed [3]. Learning complex patterns from the high-dimensional data of deep learning makes it a suitable solution for detecting network intrusion. Building detection models using machine learning or deep learning algorithms has been widely used in NIDS [4]. Although previous research has shown that they can achieve significant performance compared to traditional methods [5], their better performance usually requires large amounts of attack instance data to support. However, in the real world, there are not many high-quality samples of attack network traffic available to organizations, and most network traffic datasets suffer from serious class imbalance. These problems also exist in the widespread use of real datasets UNSW-NB15 [6], CSE-CIC-IDS2018 [7] and CICIDS'17 [7]. Specifically, the data belonging to the normal network traffic of majority class accounts for a large proportion in the dataset samples, while the data belonging to the attack network traffic of minority class accounts for a small proportion compared to the normal network traffic data of majority class. The class imbalance problem may degrade the detection performance in recognizing the attack network traffic data of minority class, leading to a large number of false positives [8].

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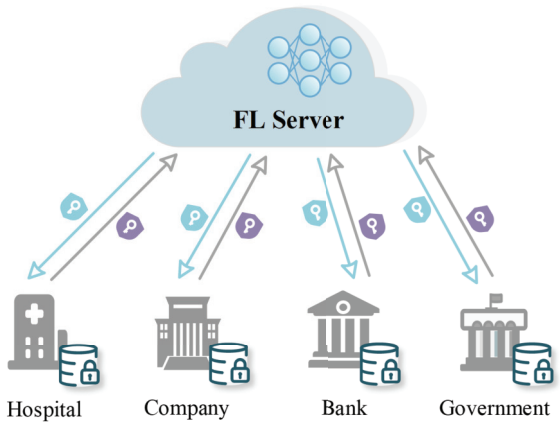

Fig. 1. An application example of federated learning, the server and various organizations collaborate to train the global model.

The intrusion detection model constructed based on a limited number of imbalanced network traffic data and deep learning methods have limitations such as insufficient ability to detect network intrusion [9]. An effective solution is to collect network traffic data from different organizations into a data center and train deep learning models based on data samples from the data center. Each organization contains a certain amount of network traffic data that exists in the form of silos. However, it is a fact that attack network traffic data is private and confidential, which involves the security of governments, enterprises, research institutes, and even the countries. In addition, there are data protection regulations and privacy policies such as GDPR and CCPA, and organizations are not allowed to share network traffic datasets. Therefore, the Federated Learning (FL) framework is needed to address this issue [10]. In the FL framework, the demand for sufficient training data can be addressed while keeping the data local to the client, and a multi-party shared global model can be collaboratively trained while ensuring client data privacy and security, as shown in Fig. 1. However, currently FL is mostly used in the field of image data, and few studies have applied it to the field of tabular data for network intrusion detection [11], [12]. Another challenge is that the distribution of network traffic data owned by each organization is imbalanced. That is, the normal network traffic data accounts for a large proportion in the dataset, and the attack network traffic data is much less than the normal network traffic data. This makes the model tend to learn more from the normal network traffic data of majority class, while ignoring the more important attack network traffic data of minority class.

In FL, the distribution of network traffic datasets for each client is usually imbalanced, and the imbalance in client dataset is a relative imbalance. That is to say, the number of attack network traffic data is not absolutely rare [13], but takes a small proportion in the whole dataset compared with the number of normal network traffic data. The number of clients participating in FL usually involves thousands or even millions, and the data varies greatly between different clients due to user habits and devices used. In other words, the degree of imbalance between normal network traffic data and attack network traffic data is not the same in the local datasets of different clients. The clients of different sizes may have different amounts of data in their network traffic datasets. In addition, due to the differences in local data distribution among clients, from a global perspective, the overall data distribution obtained by aggregating all client datasets is often imbalanced [14]. That is, attack network traffic data accounts for a small proportion in the whole global dataset, and most of the data is normal network traffic data. In this study, the imbalance problem in FL is summarized into three categories: 1) local imbalance, the local network traffic data distribution of each client is imbalanced. That is, compared with normal network traffic, the proportion of attack network traffic in the client dataset is much smaller, and the local imbalance is relative imbalance; 2) global imbalance, from a global perspective, the collection of network traffic data from all clients is imbalanced; 3) size imbalance, the number of the network traffic data owned by each client is uneven. The main goal of FL is to protect the data privacy of all clients participating in federated training and collaborate with all parties to train a shared global model with better performance [15]. However, previous studies have shown that the problem of imbalance across clients can hinder the goal and prolong the training time, and the class imbalance leads to lower classification accuracy for the attack network traffic data of minority class [16].

To solve the above problems, we propose Fed-UGI, which takes into account the data privacy protection of each organization and applies it to the learning and detection of imbalanced network traffic data. The contribution of this study can be summarized as follows:

- The proposed Fed-UGI can preprocess imbalanced datasets to construct balanced training data blocks for local models that trained with different data blocks in each local epoch, while keeping the data local to the client for privacy.
- This study proposed a new client weighted aggregation strategy based on Local Gini impurity (LG), which takes into account the differences among different client data and can reduce the impact of local models that perform poorly during training process on global model.
- Extensive experiments are conducted on real datasets, and the experimental results demonstrate that the Fed-UGI can achieve better performance of the network intrusion detection model with the advantage of reducing the training time of the model by 51.76%-92.58%, especially in highly class imbalance situation.

The rest of the paper is organized as follows. In Section II, we review the related research work, including the FL in NIDS, imbalanced data learning and the imbalanced data challenge in FL. In Section III, the motivation is elaborated. In Section IV, the Fed-UGI method proposed in this study is introduced in detail. In Section V, in order to evaluate the performance of Fed-UGI, sufficient experimental verification is performed on the basis of existing datasets. In Section VI, we set up ablation experiments to discuss the effectiveness of each improvement in the Fed-UGI. In Section VII, we discuss the limitations of our approach in this study and the solutions

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we envision to address them. The contents and experimental results of this study are summarized, and the future research work is prospected in Section VIII.

II. Related Work

A. Federated Learning in NIDS

FL, a distributed machine learning framework, can realize mobile device training on the premise of data privacy and security [17]. FL eliminates the need for data sharing as FL provides collaborative learning between devices and machine learning models can be trained on data distributed across multiple devices [18], [19]. In the FL system, server randomly selects a subset $\mathbb{K}_t \subseteq K$ of clients to participate in training in each round of communication and provides them with the current global model [20]. Clients train the downloaded global model on the local dataset and upload the updated models to the FL server to aggregate. Repeat the process many times, and eventually build a general and powerful machine learning model. FedAvg [21] is the most basic algorithm in FL, where the server aggregates the global model parameters according to the local dataset size of the client:

$$W^{t+1} = \sum_{k \in \mathrm{K}_t} \theta[k] W_k^{t+1} \tag{1}$$

where θ is the federated aggregation vector which determines the contribution of the received local models and W_k^{t+1} denotes the updated model of client k. FedAvg employs the number of local samples $| D_k |$ of client k as the federated aggregation vector, with weights proportional to the size of the local dataset:

$$\theta[k] = \frac{|D_k|}{\sum_{j \in \mathrm{K}_t} |D_j|}, \forall k \in \mathrm{K}_t. \tag{2}$$

By analyzing Eq. 2, the aggregation strategy of FedAvg is to aggregate model parameters according to the size of the client local dataset. Such aggregation method implicitly implies that models trained on more samples are better than models trained on fewer samples, thus increasing their contribution to the aggregation process. However, relying solely on the size of the data may not be reasonable, as the data distribution between clients in FL is uneven and imbalanced. This strategy ignores other important properties of the model and data, especially in terms of training convergence, which may lead to problems [22]. Considering this situation, if a client has a large but severely imbalanced dataset, then the model parameters trained on this client will differ significantly from the optimal values. However, the traditional aggregation strategy based on dataset size will make the updated global model parameters tend to the client model parameters, which will lead to the global model performance deviation from the optimal value, and even lead to the risk of global model divergence on the imbalanced dataset [23].

Recently, some aggregation methods have improved model aggregation based on FedAvg, but FedAvg is the first and most widely used. MOON [24] aims to optimize model aggregation by adding model contrast loss as a regularization term to minimize the differences between local and global models. FedAMP [25] performs personalized aggregation on both the server and the client, regardless of local targets. PartialFed [26] learns aggregation policies locally to select parameters in a global or local model. However, this choice does not accurately capture the required information in the global model. The personalization aggregation process in these methods cannot not directly applied to most existing FL frameworks and does not focus on information such as data distribution and quality of the client local dataset. These findings inspired us to further optimize the model aggregation process by using a new metric to analyze the quality of the client local dataset and fairly measure the data distribution differences among clients to determine the contribution of their local model parameters in the aggregation process.

Since researchers proposed FedAvg [21] in 2017, there is a significant development on FL, leading to FL application in a variety of fields involving sensitive data. However, few studies have applied FL to the field of network intrusion detection. Some researchers have proposed the use of FL to perform intrusion detection work in IoT networks [27]. FedDef [28], a FL-based NIDS, is an optimization-based input perturbation defense strategy with theoretical guarantees that achieves high utility by minimizing the gradient distance and strong privacy protection by maximizing the input distance. EEFED [29], a FL framework for collaborative development of security intrusion detection models, effectively reduces the negative impact of data imbalance and non-IID on FL and improves the stability of the model through a personalized updating algorithm and an optimal backtracking parameter replacement strategy.

Recently, research on network intrusion detection that applies FL to various scenarios mainly focuses on protecting privacy and reducing communication overhead [30], [31], and few studies have paid attention to the problem of declining accuracy of intrusion detection due to the imbalance of network traffic data distribution [32].

B. Imbalanced Network Traffic Data Learning

It is general that the data distribution is imbalanced in the real world. The distribution of normal traffic and attack traffic data in the network is usually imbalanced. When there is class imbalance in the data distribution, the normal traffic samples of the majority class account for a very large proportion of the whole training data, while the attack samples of the minority class will account for a much smaller proportion. Class imbalance can seriously affect the performance of machine learning detection algorithms [32], [33]. FedAvg is a classical training method that can train models and aggregate them with local datasets from collaborative clients while protecting data privacy. However, it does not take into account the impact of imbalanced data distribution on model training. The distribution of local network traffic datasets on clients is relatively imbalanced, which results in poor detection performance of the model. On the one hand, in the training process, the model will learn more from normal network traffic samples, while ignoring more important attack network traffic samples, which will cause its prediction results to be biased to normal network traffic data, thus ignoring more important attack network traffic data. On the other hand, due to the large scale of the client

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dataset, there are a large number of redundant normal network traffic samples, resulting in imbalanced training data and a long training time for the model. As a result, FedAvg may not perform well in imbalanced data scenarios. During FL training, when the model encounters class imbalance, it may lead to the generation of skewed local models during training because the model learns more from the majority class sample and ignores the minority class sample. In general, these models affect global model convergence and performance during the aggregation phase. Relevant studies have shown that the final quality of deep neural network models is determined by the first few training cycles. In the critical period, defects such as low quality or quantity of training data will lead to irreversible degradation of model performance, no matter how much additional training is conducted after this period [34]. This phenomenon was revealed in latest work of FL [35], [36], which aroused our attention to the data quality problem of imbalanced training datasets for FL models. In view of this finding, we will design a new framework based on FL to overcome the challenge of imbalanced dataset distribution in FL. While ensuring data privacy and security, we will solve the problem of imbalanced dataset distribution, reduce the size of the training dataset, and train the model on a balanced dataset. This will improve the detection performance of the model while reducing its training time.

C. Imbalanced Data Challenge in Federated Learning

The challenge of imbalanced data in FL and its impact on FL training has attracted extensive interest in the field of machine learning research. Many approaches have been proposed to address the challenge of imbalanced data in FL. From the perspective of balanced sampling, Astraea [37] introduces an additional virtual component called an intermediary to access the label distribution of the clients, and then performs data rebalancing sampling and client rearrangement training. However, approaches similar to Astraea that require collecting information about the local data class distribution of clients will lead to the potential risk of privacy leakage, and the introduced mediator increases the complexity of FL and incurs more overhead.

Some scholars have addressed the FL with class imbalance problem in terms of client selection. Wang et al. [38] proposed, FedACS, an analytic-driven client selection framework, which quantifies the category distribution heterogeneity in a FL environment based on Hoeffding inequality, and then selects clients with low data heterogeneity to form an ideal client pool based on Thompson sampling. Tang et al. [39] proposed FedCor, an FL framework constructed with an active client selection strategy based on correlation, which can effectively mitigate the accuracy degradation caused by data heterogeneity and significantly improve the convergence of FL. CriticalFL [40] identifies the Critical Learning Period (CLP) during FL training online by examining changes in the Federated Gradient Norm (FGN) metric, and then adaptively determines the number of clients participating in each training round. CriticalFL can adapt to choose more clients to participate in training in the CLP while choosing fewer clients elsewhere, while improving test accuracy while maintaining comparable or even better communication efficiency. However, in the practical application of FL, the client available in each round of communication are unenforceable, so the applicability of these strategies is limited.

In addition, FedAMP [25] is based on an attentional messaging mechanism that enables more efficient pair-to-pair collaboration between FL clients with similar data distributions without violating client data privacy. FedGR [41] proposed the imbalanced softmax function and gravitational regularizer to deal with the imbalance of sample number within the client and promote cooperation among clients to solve the label imbalance of cross-client. Ratio Loss [16] deploys a monitor on the server or client device to monitor the training data composition of each round. When the monitor continuously detects similar imbalanced components, the system mitigates the impact that has occurred by applying the loss function Ratio Loss in FL. However, the global model has learnt the imbalanced data, which has an irreversible adverse effect on the model.

Recently, FedNoRo [42] used knowledge distillation and distance-aware aggregation functions to update the federated model, and introduced logical adjustment (LA) to address data heterogeneity and class imbalance. Zhang et al. [43] designed an efficient heterogeneity sensing client sampling mechanism, namely Fed-CBS, based on a category imbalance measurement with privacy protection, which can effectively reduce the class imbalance of client packet datasets.

III. Motivation

To show the impact of imbalanced problems in training data on FL, we use the FedAvg framework to train multi-layer perceptron (MLP) based on imbalanced datasets. However, since there is no large distributed network traffic dataset, the experiment constructs a distributed dataset by resampling the UNSW-NB15 dataset [6]. The UNSW-NB15 dataset is provided by the Cybersecurity Lab at the University of New South Wales (UNSW) in Sydney. The dataset simulates the network traffic in the real network environment and contains various common network attacks and normal traffic for intrusion detection research.

A. Imbalanced Dataset

We construct four distributed UNSW-NB15 datasets: UN1, UN2, UN3, and UN4, and the detailed information is shown in Table I. UN1 and UN2 are size imbalance and global balance, the difference between them is that UN1 is local balance and UN2 is local imbalance. UN2 and UN3 are size imbalance and local imbalance, the difference between them is that UN2 is global balance while UN3 is global imbalance. In addition, UN4 has more than twice as much training data as UN3. Note that the test set is balanced, and there are no identical samples between the training sets for each client.

B. Model Architecture

The MLP model contains two fully connected layers, in which the connection between the input layer and the hidden

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TABLE I
SETTING OF DISTRIBUTED UNSW-NB15 DATASET

Dataset	Type of Imbalance			Sample size
	Size	Local	Global	Train/Test
UN1	Imbalanced	Balanced	Balanced	53000/7200
UN2	Imbalanced	Imbalanced	Balanced	53000/7200
UN3	Imbalanced	Imbalanced	Imbalanced	53000/7200
UN4	Imbalanced	Imbalanced	Imbalanced	117000/7200

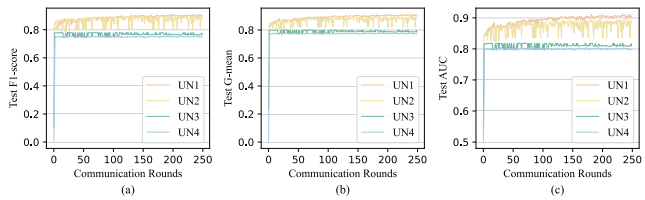

Fig. 2. F1-score, G-mean and AUC evaluated on distributed UNSW-NB15.

TABLE II
COMPARISON OF METRICS ON DISTRIBUTED UNSW-NB15 DATASET AFTER 250 COMMUNICATION ROUNDS

Dataset	Performance Metrics		
	F1-score (%)	G-mean (%)	AUC (%)
UN1	**89.87**	**90.11**	**90.15**
UN2	88.39	88.44	88.44
UN3	76.31	78.55	80.85
UN4	74.65	77.17	79.78

layer uses ReLU activation function. After activating the function, a dropout layer is created with a retention probability of 0.5 and the hidden layer is followed by a softmax output layer. In addition, the loss function is classification cross entropy, and the metrics of classification performance are F1-score, G-means and AUC, instead of accuracy or top-1 accuracy. We will show the reasons for choosing F1-score, G-means and AUC as the evaluation metrics of model performance in Section V.

C. FL Setting

We adopt a similar notation as in [17] to set the FL as follows: the communication round T is 250, the total number of clients N is 20, and the fraction C of clients participating in the communication in each round is 0.5. For local training, the local epochs E is 5 and the local batch B is 16. Each client updates the weights via the SGD optimizer, with learning rate $\eta = 0.1$ and no weight decay.

Fig. 2 shows the test results of F1-score, G-mean and AUC on four distributed UNSW-NB15 datasets. The experimental results show that local imbalance and global imbalance will lead to the degradation of model performance.

Qualitatively, for the global balance but local imbalanced dataset UN2, compared with UN1, it can be concluded from Table II and Fig. 3 (a), (b) and (c) that F1-score, G-mean and AUC decrease by 1.48%, 1.67% and 1.71%, respectively. Therefore, the local imbalance will lead to the degradation of model performance. Compared with UN2, F1-score decreased by 12.08%, G-mean decreased by 9.89%, and AUC decreased by 7.59%. For UN4, although the amount of training data of

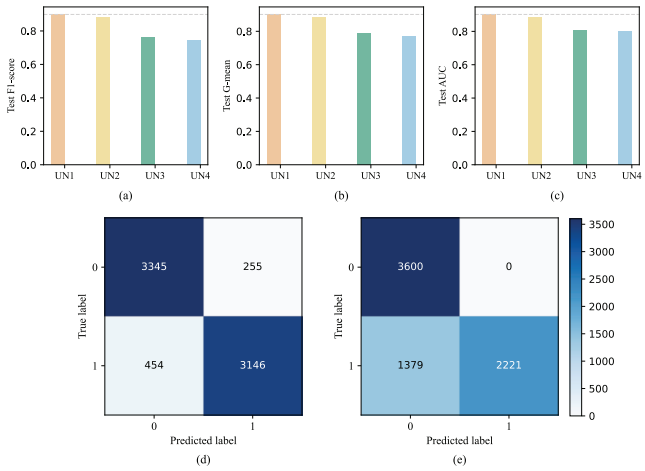

Fig. 3. Metrics and confusion matrixes evaluated on distributed UNSW-NB15 after 250 communication rounds.

UN4 is more than twice that of UN3, it does not improve the performance of the classification model.

To elucidate the impact of the class imbalance problem, Fig. 3 (d) and (e) show the confusion matrix of UN1 and UN3. Label 0 corresponds to the category of the majority class sample, and label 1 corresponds to the category of the minority class. We can find from the comparison in Fig. 3 that for the confusion matrix of UN3, category 1 (minority class) samples are not well classified, and the training process of the model is more inclined to classify the majority class samples, thus ignoring the more important minority class samples.

The class imbalance problem in FL leads to poor performance of the trained model, and the prediction of the minority class is often more difficult. More importantly, a model learning on a dataset with imbalanced distribution will lead to irreversible degradation, resulting in irreversible effects. These problems motivate us to design a method for balancing the training dataset that can preprocess the local data while protecting the client privacy, and at the same time performs more efficiently both in terms of communication and computation.

In summary, the main challenges of network intrusion detection based on FL are the class imbalanced problems in local datasets of clients and the data difference across the clients, which can lead to poor performance of local models trained by FL, and the convergence and performance of global models can be affected by the aggregation of some malicious or poorly performing local models on the server. When the distribution of the client dataset is imbalanced, the trained local model will tend to the normal network traffic data and ignore the attack network traffic data, and the parameters of the local model will deviate from the optimal value. During the aggregation phase, the local model parameters uploaded by these clients negatively affect the parameters updated by the global model, ultimately resulting in poor performance of the global model. However, it is not appropriate to upload or share the local data and data distribution information of client, as it leads to the risk of client data privacy leakage. To solve these issues, this study proposes Fed-UGI and applies it to

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imbalanced network intrusion detection. The framework firstly solves the impact of imbalanced data on local model training by preprocessing local data at the client, and improves the effect of global model aggregation through the client weighted aggregation strategy based on Local Gini impurity to further improve the classification performance and communication efficiency of the model.

Algorithm 1 Training process of Fed-UGI

Input: Total number of clients: N, total communication rounds: T, the fraction of participating clients: c, dataset of client k: D_k.

Output: Trained model W_T on round T

Clients Rebalancing
  for each client $i = 1$ **to** N **do**
    $D_i^U \leftarrow$ Algorithm 2(D_i, E);
Server Executes
  Initialize W_0;
  Collect clients Local Gini impurity: LG;
  for round $t = 1$ **to** T **do**
    $K = \max(c \cdot N, 1)$
    $\mathbb{K}_t \leftarrow$ (random subset of K clients);
      // Client Update:
      for each client $k \in \mathbb{K}_t$ parallelly **do**
        $w_{t+1}^k \leftarrow$ Algorithm 3(W_t, k);
        Send w_{t+1}^k to the server;
      end for
      // Server Update:
      $w_{t+1}^k \leftarrow$ (Collect local parameters W_{t+1}^k for $k \in \mathbb{K}_t$);
      $W_{t+1} \leftarrow$ Algorithm 4(w_{t+1}^k, LG^K);
end for

IV. Proposed Method

A. Fed-UGI Framework

In order to solve the above problems in the network intrusion detection based on FL, this study proposes Fed-UGI, which can rebalance the training dataset of each client while ensuring the privacy of the client. At the same time, the aggregation process is optimized considering the data differences between different clients. Fed-UGI is able to mitigate the imbalance problem in the FL process on both the model training and the parameter aggregation, and thus is efficient and effective in both communication and computation.

An overview of proposed Fed-UGI is shown in Fig. 4. As shown from Fig. 4, Fed-UGI consists of two components: FL server and clients. FL server is responsible for maintaining the global model, deploying the global model to the clients, and aggregating the local models uploaded by the clients. The client maintains the local model and trains the model using its local dataset. The local data distribution of the client is imbalanced, resulting in local imbalance. The data volume of different clients is different, that is, there is size imbalance.

In addition, from a global perspective, data collection is uneven across all clients and the degree of class skew varies from client to client. In the classical FedAvg, the influence of the above imbalance problem is not taken into account. Fed-UGI framework is improved on the basis of typical FL training to solve the shortcomings of existing FL methods in handling imbalanced data. Firstly, in order to solve the problem of local imbalance in the clients, we rebalance the training dataset of the client and considers the privacy protection in FL. Secondly, on the basis of federated undersampling learning framework, the aggregation strategy based on Local Gini impurity is designed to eliminate the influence of global imbalance and size imbalance on global model aggregation. Combining these two approaches, Fed-UGI can achieve a better performing global model with less training time between the clients and the server in a data privacy protected and efficient way.

Algorithm 1 shows the training process of Fed-UGI. Before FL training, client re-balancing strategy is adopted to construct balanced training data blocks by using hash-based block undersampling (HBU) method. After starting the FL training task, the server first initializes the global model to start the training. Subsequently, the FL server starts a new round of communication t and deploys the global model to all clients. Next, the server randomly selects the clients to participate in the FL model training task, and the selected clients upload the model update parameters after completing the local training task. Finally, at the end of the communication round, FL server collects parameters uploaded by clients participating in training and weights the uploaded model parameters for aggregation based on Local Gini impurity. The server then updates the global model to W_{t+1} and ends this round. The starting model for the next communication round is W_{t+1}.

B. Fed-UGI Workflow

The Fed-UGI workflow includes client rebalancing, initialization, training, and aggregation, as shown in Fig. 5.

1) Client Rebalancing: Before the initialization phase of the FL server, the clients participating in the FL model training task need to rebalance the local data with class imbalance (①), to resolve the local imbalance in the client data. The problem of local imbalance in the client data can lead to weight divergence and precision loss of the model trained on this dataset. Considering that the local network traffic dataset of the client is relative imbalanced, the undersampling method can balance its categories, and reduce the training data scale to improve the training efficiency.

In addition, due to the data distribution between clients is different, i.e., the label of minority class in the local data of different clients may be different, that is, category label c is a minority class in client A, but it is a majority class in client B.

To solve the above imbalance problems, this study proposes the Algorithm 2, that is, an undersampling method that uses hash method [44] to divide majority class samples and construct balanced data blocks with the same number of the local epoch.

The Algorithm 2 can make good use of the distribution characteristics of the dataset and retain more information of the majority class samples. The local model trained on the balanced data block can better learn the features of the minority class samples, preventing the model from ignoring the minority class samples and improving the classification

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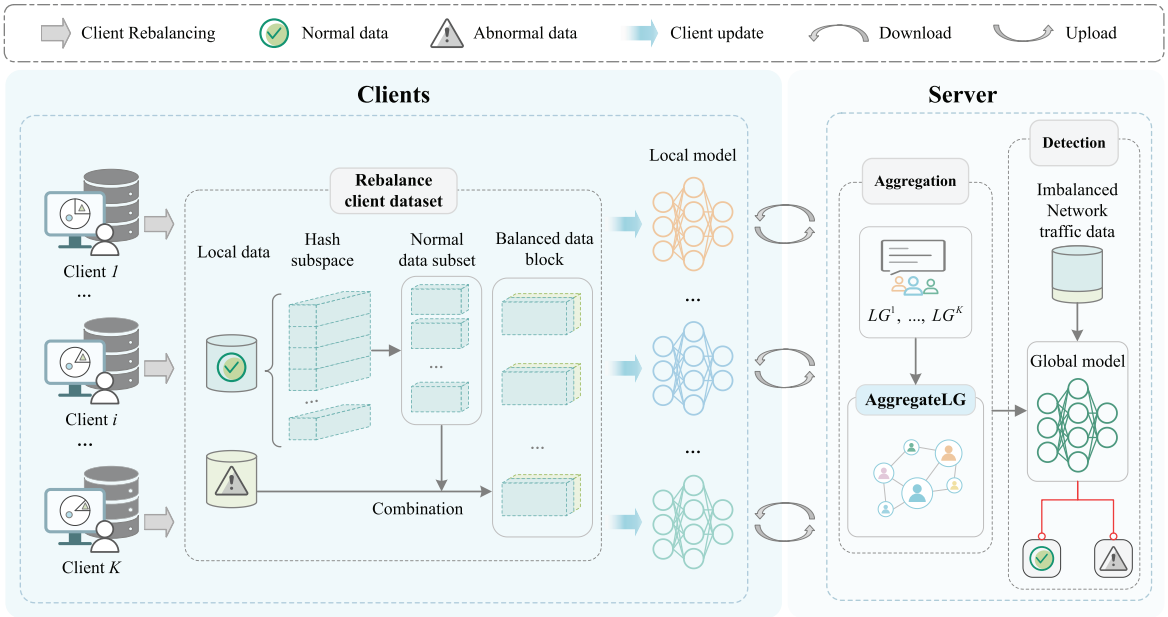

Fig. 4. Fed-UGI framework overview.

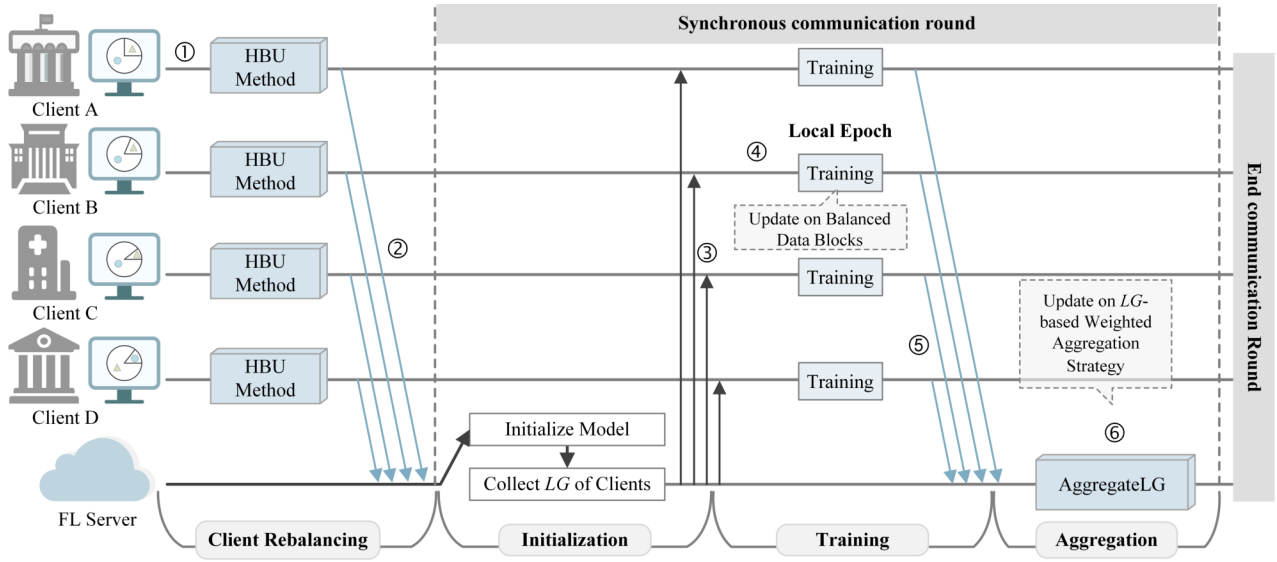

Fig. 5. Fed-UGI workflow.

performance of the model. At the same time, we also considered the need to protect data privacy, achieving a balance in the training dataset while ensuring that the data is local to the client, avoiding the risk of data leakage. The Algorithm 2 has solved the local imbalance in clients and relieved the global imbalance to a certain extent.

Due to the mismatch in class distribution between different clients, the Algorithm 2 adaptively discriminates between majority and minority classes in the dataset by counting the number of occurrences of samples from each class in the dataset (lines 1-3). Define the number of data blocks dividing the client dataset consistent with the local epoch (line 4), and define the number of hash bits as [45]:

$$N_B = \lceil \log_2 \frac{3N_{maj}}{N_{min}} \rceil \tag{3}$$

where N_{maj} and N_{min} denote the sample number of majority and minority class, respectively. Firstly, the Algorithm 2 uses the Iterative Quantization (ITQ) method to encode a binary hash for each majority class sample and divides them into hash subspaces based on the encoding. Samples with the same hash encoding will be divided into the same hash subspace (lines 7-9). ITQ [44] is an unsupervised hashing method that preserves the similarity information between the samples well. The ITQ method processes samples of majority class by principal component analysis (PCA) to find the direction of high variance of the samples in the feature space and to get their representations in a new orthogonal projection space. In this new projection space, a unit hypercube is formed centered on the origin of the PCA projection space. Each vertex of the hypercube represents a hash code, and each vertex is

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associated with the sample closest to it, which is assigned the corresponding hash code. Then, a rotation operation is performed with the aim of minimizing the quantization loss between the hash code and the original feature vector of the sample, denoted as:

$$Q(B,R) = \mathrm{P}B - VR\mathrm{P}^2 \tag{4}$$

where B, R, V and $\|\cdot\|$ denote the hash codes of the dataset samples, the rotation matrix, real-valued data descriptors after the PCA projection, and the Frobenius norm, respectively. The hash code and rotation matrix are iteratively updated to minimize quantization loss. Finally, the optimal set of hash codes and the optimal rotation matrix of the sample are obtained.

Algorithm 2 The HBU for rebalancing client datasets

Input: Client training dataset: $D = \{(x_i, y_i)\}$, $i = 1, \ldots, p$; $y_i \in \{0,1\}$, local epoch E.
Output: Processed dataset: D^U.
Clients:

1: Calculate the size of class 0 and class 1: $\mathbb{C}_0$, $\mathbb{C}_1$;
2: Identify the majority and minority class: D_{maj}, D_{min};
3: Identify the size of D_{maj} and D_{min}: N_{maj}, N_{min};
4: Identify the number of data block: $N_d \leftarrow E$;
5: Calculate the number of hash bits N_B by Equation (3);
6: Calculate the number of hash subspaces: $n = 2^{N_B}$;
7: Perform hash code on each sample in D_{maj} by using ITQ;
8: Assign each sample in D_{maj} to n hash subspaces;
9: **for** $i = 1$ to N_d **do**
10: Weighted all samples in D_{maj} by Equation (5);
11: Select N_{min} samples in D_{maj} according to weight distribution to form D^*;
12: Obtain balanced data block $d^i \leftarrow D^* \cup D_{min}$;
13: Append d^i to D^U;
14: end for
15: return D^U

Next, the Algorithm 2 constructs balanced training data blocks. Each training data block consists of all samples in the minority class and the same number of samples in the majority class as the minority class. The majority class samples are divided into hash subspaces by assigned hash codes. First, each sample i is weighted, and the sampling weight based on Hamming distance is defined as [45]:

$$w_i^H = \begin{cases} 1, & d_i = 0 \\ \dfrac{1}{n \cdot d_i}, & d_i \neq 0 \end{cases} \tag{5}$$

where n and d_i denote the number of the hash subspaces and the Hamming distance between the hash codes corresponding to sample i and each hash subspace. After that, according to the weight distribution $wH\ i$, N_{min} samples are sampled from majority class to form a majority class subset D^*, and a balance training data block is obtained by combining D^* and minority class D_{min} (line 10-12). Finally, the local dataset of client is preprocessed, and a new dataset D^U for client model training is constructed. The Algorithm 2 solves the influence of local imbalance in the training dataset and reduces the global imbalance at the same time, but it does not completely eliminate the global imbalance and size imbalance, because it is considered that the useful information of majority class may be lost when constructing data blocks with too few samples of majority class.

2) Initialization: In the initialization phase of the FL model training task, the FL server first waits for the clients participating in the training to join. Clients participate in training by sending calculated Local Gini impurity values to the server (②). Gini impurity [46] denotes a measure of the probability that a randomly selected sample in the set of samples is misclassified, which measures the degree of disorder in the set. Given that the set D contains M categories and the probability of sample points belonging to category m is p_m, then Gini impurity is defined as:

$$Gini(D) = 1 - \sum_{m=1}^{M} p_m^2 \tag{6}$$

The larger the Gini impurity value, the more chaotic the set is, the greater the uncertainty, and the greater the amount of information. On the contrary, the smaller the Gini impurity value, the more ordered the set is, the smaller the uncertainty, and the smaller the amount of information. We consider the case that the network traffic dataset has two categories in this study, i.e., normal traffic data and attack traffic data. The sample set D_k of client k can be divided into two subsets C_0 and C_1, and the Local Gini impurity of client k is defined as:

$$LG(k) = 1 - \sum_{i=0}^{1} \left(\frac{|C_i|}{|D_k|} \right)^2 \tag{7}$$

where $|D_k|$ and $|C_i|$ denote the size of dataset D_k and the size of each subset C_i. It has been analyzed that the larger the $LG(k)$ value of the local dataset D_k of client k, the higher the value of Gini impurity is, indicating that the dataset is more discrete, more random and more informative. Once the clients participating in federated training are identified, the FL server initializes the weights and optimizers of the global model and collects LG uploaded by all clients.

3) Training: In other FL frameworks, the local models of clients are trained with the same training datasets in each local epoch, while the proposed Fed-UGI training the local model with different data blocks in each local epoch. And each training block is perfectly balanced.

At the beginning of each communication round, the Server sends the current global model to all clients (③) and randomly selects fraction C clients to participate in FL training. The detailed process of client training is shown in Algorithm 3.

The client involved in the training first downloads the global model from the server. Each client performs rounds E of local training during the local training process. In each epoch, the model is trained on the balanced data block $\mathcal{D}_e$ using the mini-batch SGD algorithm (④):

$$w' = w - \eta \nabla 1(w; D_e) \tag{8}$$

Training under this framework can make the model better learn the features and distinctions between different categories and avoid the bias of sample categories. All participating clients

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Algorithm 3 The process of ClientUpdate

Input: Global model W_t on round t, local epochs: E, client k rebalanced dataset: D_k^U.
Output: Trained model w_{t+1}^k on clients k.

1: Download global model from server $w_t \leftarrow W_t$;
2: **for** each Local Epoch $e = 1$ to E **do**
3: $\mathcal{D}_e \leftarrow$ (extract balanced data block from D_k^U.);
4: $w_{t+1}^k \leftarrow w_t - \eta \nabla \ell(w_t; \mathcal{D}_e)$;
5: **end for**
6: **return** w_{t+1}^k to Server

Algorithm 4 The AggregateLG for Server update

Input: The uploaded parameters of clients: w_{t+1}^K, the Local Gini impurity of clients: LG^K.
Output: Updated model W_{t+1}.
Server:

1: **for** Each Client $k \in \mathbb{K}_t$ **do**
2: Calculate the federated aggregation vector $\theta[k]$ by Equation (9);
3: **end for**
4: Aggregate local models to W_{t+1} by Equation (10);
5: return W_{t+1}

send updates of the model to the FL server after completing the local training (⑤).

4) Aggregation: Since the data of each client are independently, the distribution quantity and probability of each category sample in the local dataset of different clients will vary greatly. So, a new client weighted aggregation strategy is proposed, which takes into account the dispersion degree and information difference of the original data from each client, and based on this, measures the contribution degree of local model parameters of each client in the aggregation process. We weight the local parameters collected by FL Server based on the Local Gini impurity uploaded by the clients for weighted aggregation (⑥), with weights proportional to the Local Gini impurity value, and define the LG-based federated aggregation vector as:

$$\theta[k] = \frac{LG(k)}{\sum_{j \in \mathrm{K}_t} LG(j)}, \forall k \in \mathrm{K}_t. \tag{9}$$

Finally, FL server aggregates all the local parameters and updates the global model to:

$$W_{t+1} = \sum_{k \in \mathrm{K}_t} \theta[k] w_{t+1}^k \tag{10}$$

The global model parameter updating process of FL server is shown in Algorithm 4, which mainly includes two steps. First, the server calculates federated aggregation vector to determine the weights of local parameter aggregation according to the Local Gini impurity of participating training clients. Next, the global parameters are calculated according to Eq. 10 to obtain the global model W_{t+1} for the next communication round.

TABLE III
EXPERIMENTAL NOTATIONS

Notation	Description
R	Communication round
η	Learning rate
B_t	Test batch size
E	Local epochs
B	Local batch size
N	Total number of clients
c	The fraction of clients participating in each training round
α_d	Dirichlet distribution coefficient

TABLE IV
DATASET INFORMATION

Dataset	Feature	Label	Class	Sample size	IR
UNSW-NB15	43	0	Normal	164673	1.77
		1	Attack	92998	
CSE-CIC-IDS2018	78	0	Normal	235776	5.10
		1	Attack	46202	
CICIDS'17	78	0	Normal	84025	38.54
		1	Attack	2180	

TABLE V
PARAMETER SETTINGS ON DIFFERENT DATASETS

Dataset	Client Parameter			Model Setup
	α_d	N	c	
UNSW-NB15	1.0	100	0.1	15
CSE-CIC-IDS2018	1.0	50	0.2	30
CICIDS'17	1.0	20	0.5	30

V. EXPERIMENT AND EVALUATION

In this section, we evaluate the performance of Fed-UGI through experiments. First of all, the detailed settings during the experiment are given, including the introduction of the datasets, the SOTA FL methods, the FL parameter settings and the evaluation criteria. Then, the performance of the proposed Fed-UGI for network intrusion detection is compared with some SOTA methods. The symbols used in the experiment are listed in Table III.

A. Experimental Settings

1) Datasets: Three widely used network intrusion detection datasets are used for experiments, namely UNSW-NB15 [6], CSE-CIC-IDS2018 [7] and CICIDS'17 [7]. For the experiments conducted on each dataset, our method and other comparison methods all adopt 10-fold cross-validation, taking the average value of ten results as a comprehensive evaluation of model performance to reduce random errors. The detailed information of the dataset used in this study, such as the number of features, label attributes, number of samples in each class and imbalance ratio (IR), are shown in Table IV.

2) Environmental Setup: In the Fed-UGI method, we use cross entropy as a loss function and set the number of communication rounds R to 250, SGD with a learning rate η to 0.1 as the optimizer for all optimization processes, and test

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TABLE VI
CONFUSION MATRIX

Confusion Matrix		Predicted	
		Attack	normal
Actual	Attack	True positives (TP)	False Negatives (FN)
	normal	False Positives (FP)	True negatives (TN)

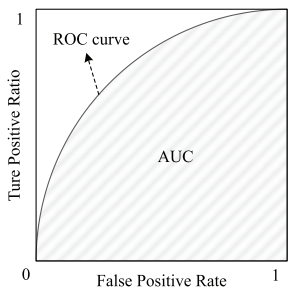

Fig. 6. ROC curve and AUC diagram.

batch size to 512. For local training, the local epoch and local batch size are set to 5 and 16, respectively. For heterogeneous data distribution among clients, Dirichlet distribution is used to divide the data. In this study, MLP is used as global model and local model, each of which has two fully connected layers, where the connection between the input layer and the hidden layer uses a ReLU activation function. The structure of the model is adjusted for different datasets with different settings for the number of hidden layer neurons, but the same structure is maintained between the comparison methods. The MLP model is implemented in PyTorch [47]. And we implement the proposed Fed-UGI based on the PyTorch federated framework [21]. The settings of client-related parameters for each dataset and the number of neurons in the hidden layer of the MLP model setup are shown in Table V. The total number of clients N is divided according to the size of the three datasets, and the value of Dirichlet parameter α_d is set to 1.0 to simulate the scenario of imbalanced client data distribution in FL. The fraction c sets 10 clients to participate in each training round for each dataset. The number of neurons in the hidden layer in the MLP model is set differently depending on the number of features in the data set but all models use the same MLP on the same data set to ensure fairness.

3) Baseline and Comparison Methods: We compare our proposed Fed-UGI with the other five SOTA baseline methods including Ratio Loss [16], FedCor [39], FedNoRo [42] and CriticalFL [40], FedDef [28]. In addition, we have added an experiment on the model performance of all data owners trained on local datasets without the FL framework.

4) Performance Evaluation Metrics: accuracy is not a suitable criterion for evaluating model performance in imbalanced data classification problems. For example, in a network traffic dataset, the ratio of normal traffic data to attack traffic data is 99:1. The model tends to predict that all data is normal, then the classification accuracy of normal traffic will reach 99%. However, the classification accuracy of attack traffic will be 0, which is not meaningful in practical applications, and it is likely to cause immeasurable losses due to misidentification of attack traffic as normal traffic. Therefore, in order to reasonably evaluate the effectiveness of our approach, we used three evaluation metrics commonly used in imbalanced data classification study, F1-score, G-mean and the area under curve (AUC) measure model performance based on the receiver operating characteristic (ROC) curve [48], [49]. In addition, we compare the training time of all methods from the start of FL training to the end of the final communication. The results of the evaluation metrics are calculated through the confusion matrix shown in Table VI:

$$F1 - score = \frac{2 \cdot \frac{TP}{TP+FN} \cdot \frac{TP}{TP+FP}}{\frac{TP}{TP+FN} + \frac{TP}{TP+FP}} \tag{11}$$

$$G - mean = \sqrt{\frac{TP}{TP+FN} \cdot \frac{TN}{TN+FP}} \tag{12}$$

The shaded area under the ROC curve in Fig. 6 is the value of AUC. and the x- and y-axes represent the false positive rate and true positive rate, respectively. The larger the value of AUC, the higher the performance of the model.

5) Statistical Tests: To further discuss and evaluate the effectiveness of Fed-UGI objectively, we used statistical tests to determine whether there are significant differences in model performance across all methods on different network intrusion detection datasets. The statistical test has been used in a number of empirical studies in machine learning and other fields [50], [51]. The statistical tests process consists of three steps. First, on each dataset, all methods are ranked from best to worst according to the test performance of each method, and the order values 1, 2, ..., the best method order value is 1; if the test performance of any of the methods is the same, the order value is equally divided. The Friedman test is then used to determine whether these methods all perform identically. If not, then the hypothesis that all methods perform identically is rejected, indicating that the performance of the methods is significantly different. A post-hoc test is needed to further distinguish the methods. Finally, Nemenyi post-hoc test is used to follow up to check whether there is a significant difference between any two methods. Nemenyi post-hoc test determines the gap between the two methods according to the critical range. If the difference between the average order values of the two methods exceeds the critical range, the hypothesis that the two methods have the same performance is rejected with the corresponding confidence level. Therefore, the test results concludes that the performance of the two methods is significantly different. In this study, the confidence level $\alpha = 0.05$.

B. Experimental Results

In this section, we experiment with Fed-UGI and five SOTA baseline methods on the UNSW-NB15, CICIDS'17 and CSE-CIC-IDS2018 datasets. In addition, the statistical test results of all methods are given.

1) Performance Comparison Results: As shown in Table VII, we give the results under the evaluation metrics of F1-score, G-mean and AUC, and the training time required for all methods to complete 250 communication rounds, respectively. The results show that Fed-UGI achieves the best

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TABLE VII
COMPARISON RESULTS OF MODEL PERFORMANCE FOR EACH METHOD

Dataset	Method	Evaluation Metrics			
		F1-score (%)	G-mean (%)	AUC (%)	Training Time (s)
UNSW-NB15	Ratio Loss	80.64±1.40	84.14±1.37	84.56±1.13	1167.16±1.54
	FedCor	42.56±28.98	21.23±31.03	56.57±12.95	1019.32±80.41
	FedNoRo	49.78±32.59	53.68±35.14	69.22±12.59	1271.31±271.43
	CriticalFL	46.56±31.62	37.94±38.11	64.03±14.31	2104.32±11.82
	FedDef	28.14±21.54	32.66±19.81	52.08±3.56	2230.95±6.22
	Fed-UGI (Ours)	**81.77±1.01**	**85.75±0.93**	**85.83±0.89**	**376.69±0.41**
CSE-CIC-IDS2018	Ratio Loss	33.79±0.91	46.41±0.81	60.19±0.36	2741.43±343.81
	FedCor	22.12±11.06	21.19±21.25	50.57±2.29	1679.08±8.27
	FedNoRo	14.27±14.29	4.11±12.33	50.50±1.49	2342.87±296.13
	CriticalFL	29.38±9.81	40.22±13.41	58.78±2.93	4127.67±390.1
	FedDef	4.50±7.20	11.43±14.97	50.27±0.76	2182.32±181.14
	Fed-UGI (Ours)	**37.22±0.96**	**53.83±0.82**	**62.03±0.47**	**810.03±157.77**
CICIDS'17	Ratio Loss	0.0±0.0	0.0±0.0	50.00±0.0	1987.02±3.45
	FedCor	2.38±3.13	6.93±20.78	49.45±9.52	1081.08±30.25
	FedNoRo	0.74±2.22	5.92±17.75	51.75±5.25	1587.45±93.56
	CriticalFL	0.0±0.0	0.0±0.0	50.00±0.0	2472.35±18.08
	FedDef	3.05±7.96	6.72±12.98	50.64±2.69	2569.94±218.72
	Fed-UGI (Ours)	**67.42±6.25**	**96.52±1.27**	**96.55±1.24**	**183.57±25.64**

results on all metrics. On the UNSW-NB15 dataset, compared with Ratio Loss that performs best in the baseline methods, our method increases by 1.13%, 1.61% and 1.27% in F1-score, G-mean and AUC, respectively, and more importantly, the training time is reduced by about 67.73%. On the CSE-CIC-IDS2018 dataset, compared with the optimal Ratio Loss, Fed-UGI further improves by 3.43%, 7.42% and 1.8%, respectively. Compared with other baseline methods, the training time is reduced by approximately 51.76% to 80.38%. On the CICIDS'17 dataset, the results of all the baseline methods are poor, which indicates that none of these methods can effectively identify attack traffic samples of minority classes in the network traffic data, or even ignore the attack traffic samples altogether in the highly imbalanced situation.

The model obtained from Fed-UGI training can achieve the best detection results in terms of F1-score, G-mean and AUC, which can effectively solve the adverse effect of highly imbalanced data on model training and reduce the training time by 83.02%-92.86% compared with other methods. Based on this, it can be seen from the experimental results that our method achieves the optimal detection effect and can effectively detect normal and attack network traffic data in imbalanced network traffic data, especially in highly class imbalance situation. Meanwhile, the training time required is drastically reduced to improve the training efficiency of the model.

In the experimental results on the UNSW-NB15, CICIDS '17 and CSE-CIC-IDS2018 datasets, the Fed-UGI method performed better in detection performance on the UNSW-NB15 and CICIDS' 17 datasets. It is considered that the three datasets are different in terms of data scale, degree of imbalance in data distribution and number of features. Our Fed-UGI first uses HBU method to balance the data distribution of the highly imbalanced local dataset on the client, and obtains the balanced dataset to train the local model, while reducing the dataset size and training time. However, the local model of other baseline methods is trained directly on the highly imbalanced dataset, and the obtained model parameters will seriously deviate from the optimal value. Although these baselines will later improve model performance through loss functions and knowledge distillation, the impact of highly imbalanced data on the model is irreversible, and in the aggregation stage, these deviated model parameters will exacerbate the divergence of global model parameters, leading to poor performance of baseline methods. In addition, LG-based weighted aggregate strategy proposed in Fed-UGI can further optimize the effect of model parameter aggregation. As a result, our Fed-UGI has the best performance over other baseline methods on F1-score, G-mean and AUC, and shows a definite advantage in training time.

In Fig. 7, the detection performance of models trained by all data owners on their local datasets is shown. As shown in Fig. 7 (a), on the UNSWNB15 dataset, the dotted lines in the figure represent the performance results of our Fed-UGI method. It can be found that without FL framework, the performance of 100 data owners on F1 score, G-mean, and AUC metrics is poor. In Fig. 7 (b) and (c), for the CSE-CIC-IDS2018 and CICIDS '17 datasets, the values of F1-score and G-mean for the model trained by all data owners on imbalanced local datasets are 0, and the values of AUC are 0.5, indicating that the model cannot identify attack network traffic samples. All samples are predicted as normal network traffic samples, and the model tends to predict randomly. Fed-UGI is based on a FL framework that collaborates with various clients for training which can learn sufficient attack network traffic data and normal network traffic data.

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TABLE VIII
ORDER VALUE OF METHODS COMPARISON ON EACH METRIC

Metric	Dataset	Method					
		Ratio Loss	FedCor	FedNoRo	CriticalFL	FedDef	Fed-UGI (Ours)
F1-score	UNSW-NB15	2	5	3	4	6	1
	CSE-CIC-IDS2018	2	4	5	3	6	1
	CICIDS'17	5.5	3	4	5.5	2	1
	Average Order Value	3.167	4	4	4.167	4.667	1
G-mean	UNSW-NB15	2	6	3	4	5	1
	CSE-CIC-IDS2018	2	4	6	3	5	1
	CICIDS'17	5.5	2	4	5.5	3	1
	Average Order Value	3.167	4	4.333	4.167	4.333	1
AUC	UNSW-NB15	2	5	3	4	6	1
	CSE-CIC-IDS2018	2	4	5	3	6	1
	CICIDS'17	4.5	6	2	4.5	3	1
	Average Order Value	2.833	5	3.333	3.833	5	1
Training time	UNSW-NB15	3	2	4	5	6	1
	CSE-CIC-IDS2018	5	2	4	6	3	1
	CICIDS'17	4	2	3	5	6	1
	Average Order Value	3	2	3.667	5.333	5	1

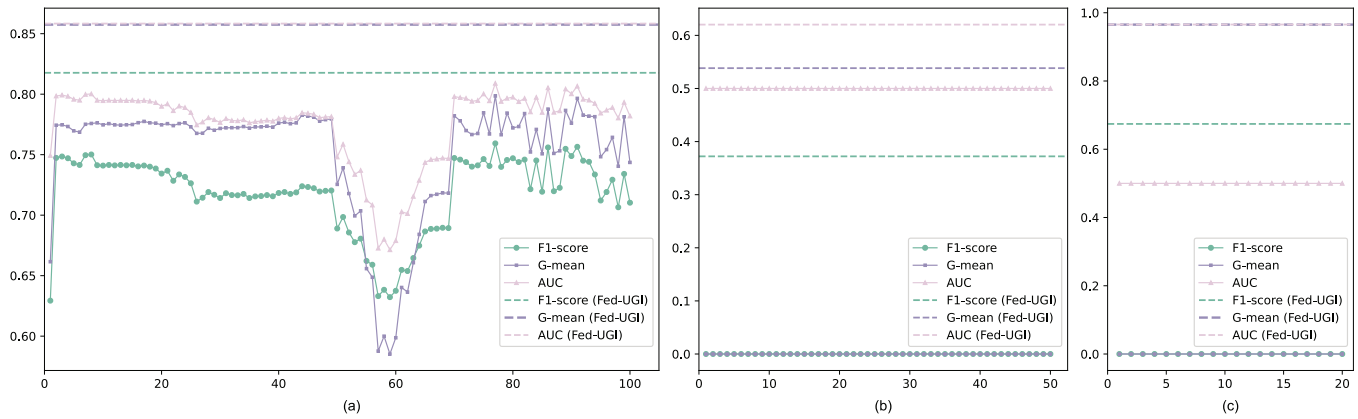

Fig. 7. Without FL framework result.

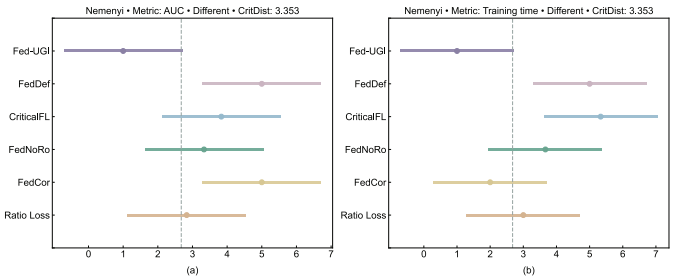

Fig. 8. Results of Nemenyi post-hoc test.

In addition, Fed-UGI solves the problem of imbalanced data distribution and avoids the skew of the model on imbalanced datasets.

2) Statistical Test Results: We used statistical test to further validate the significance of the difference in detection performance between our method and SOTA methods. For each evaluation metric, the test performance of each method on each dataset is ranked from best to worst. Table VIII shows the order value and average order value of the test performance for each method on all datasets. In terms of evaluation metrics F1-score, G-mean, AUC and training time, the results of Friedman test $\mathcal{T}_F$ are calculated to be 1.962, 1.865, 3.676, 9.250, respectively, and the critical value of F test when confidence $\alpha = 0.05$ is 3.326. In metrics F1-score and G-mean, the results of Friedman test $\mathcal{T}_F$ are all smaller than the critical value of F test, so the hypothesis that the performance of all algorithms on F1-score and G-mean is the same cannot be rejected.

On both metric AUC and training time, the Friedman test results $\mathcal{T}_F$ are greater than the critical value of F test, so the hypothesis that the performance of all algorithms is the same is rejected, indicating that the performance of all methods is significantly different on AUC and training time. The Nemenyi post-hoc test is then used to further distinguish the methods in metrics AUC and training time. Nemenyi post-hoc test calculated that the critical range *CD* is 3.353. We mainly analyzed the gap between Fed-UGI and other methods, and the test results can be visually displayed in Fig. 8, where the vertical axis shows each method, the horizontal axis is the average orderh value. For each method, a dot is used

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TABLE IX
ABLATION EXPERIMENT RESULTS

Dataset	Combination	Rebalancing	Aggregation		Evaluation Metrics			
		HBU	AVG	LGW	F1-score (%)	G-mean (%)	AUC (%)	Training Time (s)
UNSW-NB15	(1)		✓		79.18±0.57	81.73±1.00	83.01±0.49	849.37±2.08
	(2)			✓	79.47±0.81	83.70±0.82	83.90±0.73	845.87±1.99
	(3)	✓	✓		81.51±0.79	84.19±1.00	84.89±0.73	376.86±0.43
	(4)	✓		✓	**81.77±1.01**	**85.75±0.93**	**85.83±0.89**	**376.69±0.41**
CSE-CIC-IDS2018	(1)		✓		29.56±9.87	40.42±13.48	58.84±2.96	1911.21±4.59
	(2)			✓	34.08±1.97	47.97±3.78	60.45±0.96	1901.98±7.26
	(3)	✓	✓		37.11±0.84	53.71±1.21	61.98±0.47	997.31±17.81
	(4)	✓		✓	**37.22±0.96**	**53.83±0.82**	**62.03±0.47**	**810.03±157.77**
CICIDS'17	(1)		✓		0.0±0.00	0.0±0.00	50.00±0.00	1434.51±2.38
	(2)			✓	**84.82±1.64**	91.18±1.15	91.55±1.06	1436.25±0.88
	(3)	✓	✓		68.93±3.35	94.52±1.00	94.59±0.97	184.56±26.71
	(4)	✓		✓	67.42±6.25	**96.52±1.27**	**96.55±1.24**	**183.57±25.64**

to display its average order value, and the horizontal line centered on the dot represents the size of the critical range. As shown in Fig. 8 (a), in evaluation metric AUC, Fed-UGI performs better than other methods and is significantly better than FedCor and FedDef method. For metric training time shown in Fig. 8 (b), Fed-UGI outperforms other methods and significantly outperforms the CriticalFL and FedDef methods.

In conclusion, the proposed Fed-UGI can greatly reduce the training time consumption of FL model while improving the performance of network intrusion detection model. Meanwhile, the Fed-UGI outperforms and significantly outperforms the FedCor method, the CriticalFL method and FedDef method in the statistical tests of the evaluation metrics AUC and the training time.

VI. DISCUSSION

In this section, we further used the UNSW-NB15, CICIDS'17 and CSE-CIC-IDS2018 datasets for ablation experiments to verify the effectiveness of each improvement in Fed-UGI. The component AVG in the experiment refers to the aggregation strategy in FedAvg, and LGW represents the LG-based weighted aggregate strategy proposed in Fed-UGI.

Table IX shows a comparison of combining the HBU method and the LGW aggregate strategy with only one of the improved combinations on the three experimental datasets. We also implemented the FedAvg method and compared it to Fed-UGI. The combination (1) of each dataset indicates that there is no improved component, that is, the underlying FedAvg method. Combination (2) contains only LGW improvement, combination (3) contains only HBU improvement, and combination (4) contains all improvements at the same time.

The results of four groups of experiments on the dataset UNSW-NB15 are analyzed as follows. Compared with (1), after adding only the LGW aggregate strategy in (2), F1-score, G-mean and AUC increase by about 0.29%, 1.97% and 0.89%, respectively. (3) Compared with (1), which adds the HBU method, F1-score, G-mean and AUC increase by about 2.33%, 2.46% and 1.88%, respectively. When the same parameter aggregation strategy is added, (3) compared with (1), the training time is reduced by 55.63%; (4) Compared with (2), the training time decreased by 55.47%. (4) Contains HBU and LGW aggregate strategy, its performance is the best compared to (1), (2), and (3).

The results of four groups of experiments on the dataset CSE-CIC-IDS2018 are analyzed as follows. (2) Compared with (1), LGW aggregate strategy can increase F1-score, G-mean and AUC by 4.52%, 7.55% and 1.61%, respectively. (3) Compared with (1), HBU method can increase F1-score, G-mean and AUC by 7.55%, 13.29% and 3.14% respectively. Containing HBU and LGW aggregate strategy in (4), the model performs best with the least training time.

The imbalance ratio of dataset CICIDS'17 is as high as 38.54, and the client local data after Dirichlet partition is also high imbalanced. Combination (1), that is, FedAvg method in the case of highly class imbalance situation, the values of F1-score and G-mean are 0, and the value of AUC is 0.5, indicating that the model cannot identify attack samples of minority class, and all samples are predicted as normal traffic, which means that the model tends to random prediction. The results of (2) and (3) show that both HBU method and LGW aggregate strategy can solve the problem that the model cannot identify attack samples of minority class due to highly class imbalance in the local data of the client. (4) Compared with (2), under the same parameter aggregation strategy, combined with HBU method, G-mean and AUC can be increased by 5.34% and 5.00% respectively, and the training time can be reduced by 87.22%. (4) Compared with (3), LGW improvement is further added while HBU method is also adopted, which further improves the performance of the model.

The reason why the performance of the model can be improved while the training time can be reduced is that the HBU method can construct balanced training data blocks to support the model training on the premise of retaining the information of important samples of majority class, and at the same time reduce the scale of training data, thus reducing the training time. All improvements are included in (4), resulting

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in the best value. According to the results of ablation experiments, the components in Fed-UGI can effectively improve the performance of the final model detection results. Based on this, we can see that every improvement in Fed-UGI is useful. The HBU method in Fed-UGI mainly solves the local imbalance in the client, so that the local model can be trained on the balanced data and the data scale can be reduced at the same time. However, there are still differences in the degree of data dispersion and the amount of information between the clients. Therefore, it is important to combine the two strategies. After the local model training is completed, we adopted a new parameter weighted aggregation strategy to further optimize the effect of parameter aggregation, so as to achieve the goal of improving the performance of the federated model and shortening the training time. Compared with the FedAvg method, the training time is reduced by 55.65%-87.20%.

VII. Limitation

This study focuses on the binary problem of network intrusion detection. The model detects normal network traffic and attack network traffic, without considering to distinguish specific types of attacks. Each client dataset typically focuses on detecting some type of attack. In other words, the types of attacks in different client networks may not be the same. When a client uses the global model to detect attacks on its network, how to make the global model effectively detect specific attacks on the client network is worth further research. At present, we propose two solutions to solve this problem. One is to use a new detection method to further identify the attack types of all attack network traffic data based on the binary classification of network intrusion we have done. The other is to take this problem as a multi-classification problem of network intrusion detection, and identify various types of attack categories by improving the detection model. In addition, this study is to analyze historical network traffic data. For real time applications, we have not yet tried to study real-time network traffic data and we believe that the Fed-UGI can be implemented based on Apache Flink to solve this issue.

VIII. Conclusion and Future Work

Due to the continuous opening of the Internet, the security risks and threats it faces are increasing, and the effective detection of network intrusion to maintain a secure network environment is becoming more and more important. This study proposes a novel federated undersampling learning framework with Gini impurity, namely Fed-UGI. First of all, since the application of machine learning algorithms to network intrusion detection requires a large number of high-quality data to support model training, but all parties cannot share data in the security and privacy requirements of network traffic data. Therefore, it is proposed to coordinate multi-participant collaborative training in FL framework to adequately train the global model shared by multiple parties while satisfying privacy protection and data security. Secondly, we design a federated undersampling learning framework, which avoids training on network traffic data with class imbalance to produce biased models and prevents models from ignoring important attack network traffic data of minority class. By rebalancing the local training data of each client, the local model can be trained on different balanced data blocks in each local epoch, which improves the detection performance of attack network traffic data and maintains stable performance in highly class imbalance situations. Finally, based on the federated undersampling learning framework, we propose a new client weighted aggregation strategy, which measures the dispersion degree and information difference of the client data through the Local Gini impurity, to measure the contribution of local model parameters to the global model and further improve the detection performance of the global model. In addition, experimental results show that Fed-UGI can achieve higher detection performance with less training time compared to baseline methods. Finally, the ablation experiments verified that the improvements of the components in Fed-UGI are effective. It is worth noting that we provide a FL framework that is not limited by any detection model, that is, the imbalanced treatment of client rebalancing in this framework can support for any more advanced intrusion detection model, and the client weighted aggregation strategy is also generic.

In the future, we will make the framework more flexible and versatile to be applied in more critical areas. Consider that there is less data in some organizations, while there are also local imbalance, global imbalance, and size imbalance in FL. In addition, there is also the problem of multi-classification in FL based network intrusion detection, and the global model needs to detect specific attack types on the clients. In future studies, we aim to further improve the performance of the model on the basis of solving these problems without violating privacy protection requirements.

References

[1] V. Kumar and D. Sinha, "Synthetic attack data generation model applying generative adversarial network for intrusion detection," *Comput. Secur.*, vol. 125, Feb. 2023, Art. no. 103054.

[2] K. He, D. D. Kim, and M. R. Asghar, "Adversarial machine learning for network intrusion detection systems: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 538–566, 1st Quart., 2023.

[3] H. Jmila and M. I. Khedher, "Adversarial machine learning for network intrusion detection: A comparative study," *Comput. Netw.*, vol. 214, Sep. 2022, Art. no. 109073.

[4] N. Wang, Y. Chen, Y. Xiao, Y. Hu, W. Lou, and Y. T. Hou, "MANDA: On adversarial example detection for network intrusion detection system," *IEEE Trans. Depend. Sec. Comput.*, vol. 20, no. 2, pp. 1139–1153, Mar. 2023.

[5] T. Zebin, S. Rezvy, and Y. Luo, "An explainable AI-based intrusion detection system for DNS over HTTPS (DoH) attacks," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 2339–2349, 2022.

[6] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6.

[7] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. 4th Int. Conf. Inf. Syst. Secur. Privacy*, Jan. 2018, pp. 108–116.

[8] A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, "A theoretical analysis of the learning dynamics under class imbalance," in *Proc. 40th Int. Conf. Mach. Learn.*, Jul. 2023, pp. 10285–10322.

[9] J. L. Guerra, C. Catania, and E. Veas, "Datasets are not enough: Challenges in labeling network traffic," *Comput. Secur.*, vol. 120, Sep. 2022, Art. no. 102810.

[10] X. Hao, C. Lin, W. Dong, X. Huang, and H. Xiong, "Robust and secure federated learning against hybrid attacks: A generic architecture," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1576–1588, 2024.

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[11] D. Wagner et al., "United we stand: Collaborative detection and mitigation of amplification DDoS attacks at scale," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2021, pp. 970–987.

[12] E. M. Campos et al., "Evaluating federated learning for intrusion detection in Internet of Things: Review and challenges," *Comput. Netw.*, vol. 203, Feb. 2022, Art. no. 108661.

[13] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263–1284, Sep. 2009.

[14] Z. Shen, J. Cervino, H. Hassani, and A. Ribeiro, "An agnostic approach to federated learning with class imbalance," in *Proc. 10th Int. Conf. Learn. Represent.*, Apr. 2022, pp. 1–12.

[15] F. Lai, X. Zhu, H. V. Madhyastha, and M. Chowdhury, "Oort: Efficient federated learning via guided participant selection," in *Proc. 15th USENIX Symp. Operating Syst. Des. Implement. (OSDI)*, Jul. 2021, pp. 19–35.

[16] L. Wang, S. Xu, X. Wang, and Q. Zhu, "Addressing class imbalance in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2021, pp. 10165–10173.

[17] Q. Li et al., "A survey on federated learning systems: Vision, hype and reality for data privacy and protection," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3347–3366, Apr. 2023.

[18] Z. Pan, L. Hu, W. Tang, J. Li, Y. He, and Z. Liu, "Privacy-preserving multi-granular federated neural architecture search—A general framework," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 3, pp. 2975–2986, Mar. 2023.

[19] W. Huang, J. Liu, T. Li, T. Huang, S. Ji, and J. Wan, "FedDSR: Daily schedule recommendation in a federated deep reinforcement learning framework," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3912–3924, Apr. 2023.

[20] H. Zhou, G. Yang, H. Dai, and G. Liu, "PFLF: Privacy-preserving federated learning framework for edge computing," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 1905–1918, 2022.

[21] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Statist.*, Apr. 2017, pp. 1273–1282.

[22] U. Michieli and M. Ozay, "Are all users treated fairly in federated learning systems?," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 2318–2322.

[23] G. Wang, C. X. Dang, and Z. Zhou, "Measure contribution of participants in federated learning," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2019, pp. 2597–2604.

[24] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10713–10722.

[25] Y. Huang et al., "Personalized cross-silo federated learning on non-IID data," in *Proc. AAAI Conf. Artif. Intell., May.*, May 2021, pp. 7865–7873.

[26] B. Sun, H. Huo, Y. Yang, and B. Bai, "PartialFed: Cross-domain personalized federated learning via partial initialization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2021, pp. 23309–23320.

[27] P. Tian, Z. Chen, W. Yu, and W. Liao, "Towards asynchronous federated learning based threat detection: A DC-Adam approach," *Comput. Secur.*, vol. 108, Sep. 2021, Art. no. 102344.

[28] J. Chen, Y. Zhao, Q. Li, X. Feng, and K. Xu, "FedDef: Defense against gradient leakage in federated learning-based network intrusion detection systems," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4561–4576, 2023.

[29] X. Huang, J. Liu, Y. Lai, B. Mao, and H. Lyu, "EEFED: Personalized federated learning of execution & evaluation dual network for CPS intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 41–56, 2023.

[30] J. Gao et al., "Secure aggregation is insecure: Category inference attack on federated learning," *IEEE Trans. Depend. Sec. Comput.*, vol. 20, no. 1, pp. 147–160, Jan. 2023.

[31] J. Zhou et al., "A differentially private federated learning model against poisoning attacks in edge computing," *IEEE Trans. Depend. Sec. Comput.*, vol. 20, no. 3, pp. 1941–1958, Jun. 2023.

[32] H. Ding, Y. Sun, N. Huang, Z. Shen, and X. Cui, "TMG-GAN: Generative adversarial networks-based imbalanced learning for network intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 19, pp. 1156–1167, 2024.

[33] A. Ito, K. Saito, R. Ueno, and N. Homma, "Imbalanced data problems in deep learning-based side-channel attacks: Analysis and solution," *IEEE Trans. Inf. Forensics Security*, vol. 16, pp. 3790–3802, 2021.

[34] S. Jastrzebski et al., "Catastrophic Fisher explosion: Early phase Fisher matrix impacts generalization," in *Proc. 38th Int. Conf. Mach. Learn.*, Jul. 2021, pp. 4772–4784.

[35] G. Yan, H. Wang, and J. Li, "Seizing critical learning periods in federated learning," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 8788–8796.

[36] G. Yan, H. Wang, X. Yuan, and J. Li, "DeFL: Defending against model poisoning attacks in federated learning via critical learning periods awareness," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2023, pp. 10711–10719.

[37] M. Duan, D. Liu, X. Chen, R. Liu, Y. Tan, and L. Liang, "Self-balancing federated learning with global imbalanced data in mobile systems," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 1, pp. 59–71, Jan. 2021.

[38] Z. Wang, Y. Zhu, D. Wang, and Z. Han, "FedACS: Federated skewness analytics in heterogeneous decentralized data environments," in *Proc. IEEE/ACM 29th Int. Symp. Quality Service (IWQOS)*, Jun. 2021, pp. 1–10.

[39] M. Tang et al., "FedCor: Correlation-based active client selection strategy for heterogeneous federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10102–10111.

[40] G. Yan, H. Wang, X. Yuan, and J. Li, "CriticalFL: A critical learning periods augmented client selection framework for efficient federated learning," in *Proc. 29th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2023, pp. 2898–2907.

[41] S. Guo et al., "FedGR: Federated learning with gravitation regulation for double imbalance distribution," in *Proc. Int. Conf. Database Syst. Adv. Appl.*, Apr. 2023, pp. 703–718.

[42] N. Wu, L. Yu, X. Jiang, K.-T. Cheng, and Z. Yan, "FedNoRo: Towards noise-robust federated learning by addressing class imbalance and label noise heterogeneity," in *Proc. 32nd Int. Joint Conf. Artif. Intell.*, Aug. 2023, pp. 4424–4432.

[43] J. Zhang et al., "Fed-CBS: A heterogeneity-aware client sampling mechanism for federated learning via class-imbalance reduction," in *Proc. 40th Int. Conf. Mach. Learn.*, Jul. 2023, pp. 41354–41381.

[44] Y. Gong, S. Lazebnik, A. Gordo, and F. Perronnin, "Iterative quantization: A procrustean approach to learning binary codes for large-scale image retrieval," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 12, pp. 2916–2929, Dec. 2013.

[45] W. W. Y. Ng, S. Xu, J. Zhang, X. Tian, T. Rong, and S. Kwong, "Hashing-based undersampling ensemble for imbalanced pattern classification problems," *IEEE Trans. Cybern.*, vol. 52, no. 2, pp. 1269–1279, Feb. 2022.

[46] S. Nembrini, I. R. König, and M. N. Wright, "The revival of the Gini importance?," *Bioinformatics*, vol. 34, no. 21, pp. 3711–3718, Nov. 2018.

[47] A. Paszke et al., "Automatic differentiation in PyTorch," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), Workshop Autodiff Decis. Program Chairs*, 2017, pp. 1–4.

[48] Y. Yan, Y. Zhu, R. Liu, Y. Zhang, Y. Zhang, and L. Zhang, "Spatial distribution-based imbalanced undersampling," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6376–6391, Jun. 2023.

[49] T. Zhu, X. Liu, and E. Zhu, "Oversampling with reliably expanding minority class regions for imbalanced data learning," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6167–6181, Jun. 2023.

[50] J. M. Leski, R. Czabanski, M. Jezewski, and J. Jezewski, "Fuzzy ordered *c*-means clustering and least angle regression for fuzzy rule-based classifier: Study for imbalanced data," *IEEE Trans. Fuzzy Syst.*, vol. 28, no. 11, pp. 2799–2813, Nov. 2020.

[51] L. Li, H. He, and J. Li, "Entropy-based sampling approaches for multi-class imbalanced problems," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 11, pp. 2159–2170, Nov. 2020.

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